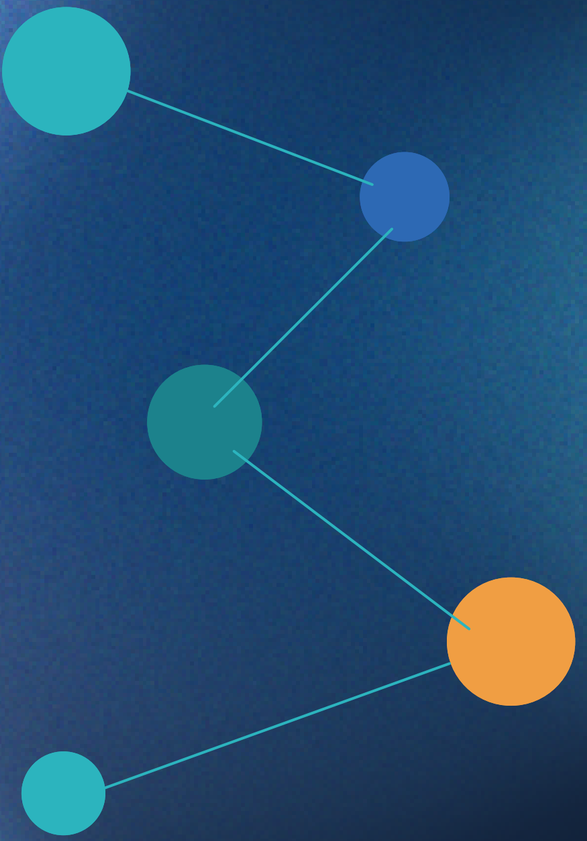
TenderLens

AI-Powered Tender Qualification & Disqualification Engine



From fragmented tender documents to evidence-backed Bid / No-Bid decisions.

DISCOVER • VERIFY • DECIDE

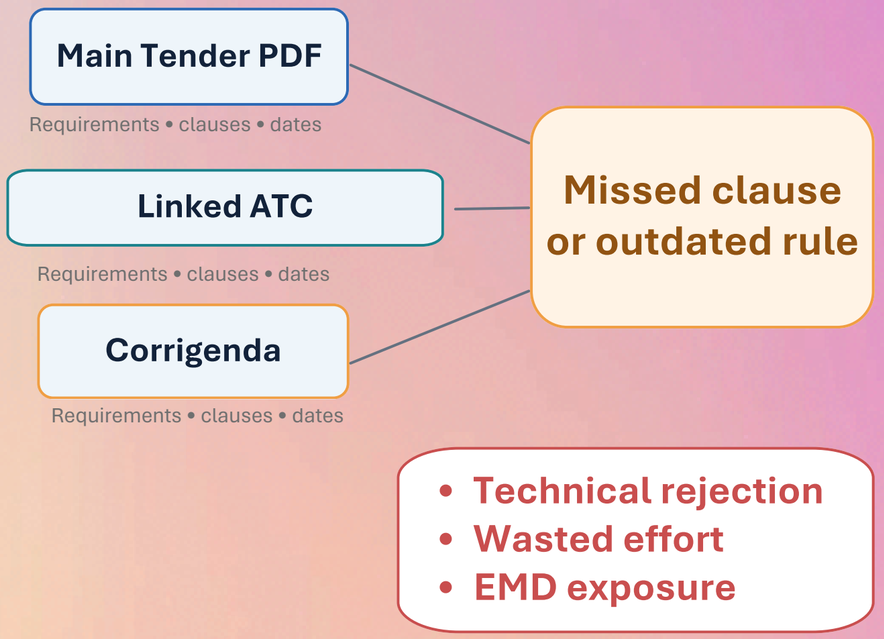
Presented by Team Traxo

Proposed system • Human review for ambiguous cases

1. Problem Statement

Finding tenders is easy. Proving eligibility is hard.

FRAGMENTED TENDER INFORMATION



THE CORE PAIN

- Hidden eligibility conditions
- Mandatory certificates in linked files
- Turnover / exemption traps
- Portal access friction

"The challenge isn't finding a tender. It's knowing whether you qualify before you bid."

THE ACTUAL PROBLEM

Indian MSMEs often struggle to verify their eligibility for government tenders because requirements are scattered across main tender PDFs, linked ATCs, and corrigenda.

- Fragmented Information: Important conditions are spread across multiple documents.
- Complex Eligibility Rules: Turnover exemptions and bidder-specific conditions can be misunderstood.
- Missing Requirements: Mandatory certifications or OEM documents may be overlooked.
- Financial Risks: Incorrect bids can cause technical rejection, wasted preparation effort, and capital locked in EMD deposits.

Need: one auditable workflow to check the complete document set before a bid is prepared.

2. Proposed Solution

Turn scattered tender documents into an evidence-backed qualification report.

Every material risk links to its source document, page and clause.

Decision support — not an automatic legal verdict. Ambiguous or conflicting cases go to human review.

STEP 01
Bidder Profile

Create a structured business profile for eligibility matching.

- **Entity type** — OEM / Manufacturer / Trader / Reseller
- **Annual turnover** (3 years)
- **Certifications** (BIS, ISO, OEM authorization)
- **Relevant experience** and past projects
- **Other details** (state, category, etc.)

Your Business Profile

Entity Type	Manufacturer
Annual Turnover (₹)	12 Crore
Certifications	ISO 9001 × BIS ×
Years of Experience	8

STEP 02
Tender Discovery

Find relevant tenders from government procurement portals using TinyFish Search.

- Discover relevant tenders based on bidder profile
- Filter by sector, location, value and eligibility fit
- Capture source URLs and available metadata
- Prioritize high-fit opportunities

Government Portals

GeM · CPPP · PSU Portals · State Portals

STEP 03
Deep Ingestion

Retrieve and process the complete tender document set.

- Main tender PDF
- Linked ATCs and corrigenda
- Extract text and tables
- Preserve page references and document versions



STEP 04
Hybrid Verification Engine

Combine deterministic rules with AI-based clause analysis.

- **Python rules for explicit requirements** (turnover, experience, numerical thresholds, certifications)
- LLM analysis for complex clauses and exceptions
- Identify potential disqualifiers and conflicts
- Flag ambiguous cases for human review

Example Checks

- Turnover ≥ required threshold
- Required years of experience
- BIS/ISO certificate present
- OEM authorization required
- Special conditions / exemptions
- Conflicting clauses across documents

STEP 05
Qualification Scorecard

Generate an evidence-backed report to support Bid/No-Bid decision.

- **Strong Bid** — Meets key requirements
- **Conditional Bid** — Gaps or clarifications needed
- **Red Flag** — High risk of disqualification

TenderLens Report — Score: 4.2 / 5.0

Eligibility Summary

Turnover Requirement	Meets (Pg 8, Cl 4.1)
Experience Criteria	Meets (Pg 12, Cl 3.2)
BIS Certification	Not found
OEM Authorization	Meets (Pg 15, Cl 5.1)
EMD Requirement	Not satisfied (Pg 7, Cl 2.3)

Source Document

"The bidder shall have an annual turnover of minimum ₹10 Crore in the last 3 financial years."

Page 8, Clause 4.1

1. Target Users

TenderLens is built for businesses and professionals who participate in government procuerment and need a reliable way to verify eligibility, reduce risk and make confident Bid/No-Bid decisions.



MSMEs & OEMs

Verify turnover requirements, exemptions and local-content conditions.

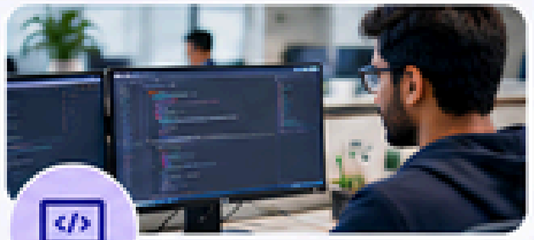
- Check turnover eligibility
- Confirm MII / local content rules
- Verify required certifications (BIS/ISO, etc.)



EPC Contractors

Validate past-project experience and technical qualification thresholds.

- Check project experience criteria
- Verify technical & financial capacity
- Ensure all mandatory documents are available



IT Companies & System Integrators

Check tender scope, certifications and OEM authorization requirements.

- Verify scope and technical specs
- Confirm OEM authorization letters
- Check software/hardware certifications



Bid Consultants

Automate eligibility checks and compliance assessments across multiple clients.

- Handle multiple tenders efficiently
- Generate client-wise reports
- Reduce manual effort and errors

PRIMARY FOCUS

Indian MSMEs

We aim to provide an affordable and reliable way for Indian MSMEs to evaluate tender eligibility before investing time and money in a bid.

OUR GOAL

Empower every eligible business to bid with confidence.

4. Technical Architecture & Stack

Modular components separate retrieval, verification and reporting.

USER EXPERIENCE
Frontend (Web Application)

- Next.js
- TypeScript
- Tailwind CSS

API & ORCHESTRATION
Backend & Orchestration

- FastAPI
- Pydantic
- Celery (async jobs)

WEB INTELLIGENCE
Search & Data Retrieval

- SerpAPI (Search API)
- HTTPX (Web requests)
- TinyFish

DOCUMENT INTELLIGENCE
Document Processing

- PyMuPDF (PDF processing)
- Docling (Structured extraction)

QUALIFICATION ENGINE
Rules & AI Processing

- Python (Business rules)
- LLM (Structured outputs)
- Rules Engine (Validation & scoring)

DATA & OPERATIONS
Database, Storage & Infra

- PostgreSQL (Relational DB)
- Object Storage (e.g., S3)
- Redis (Cache)
- Docker (Deployment)

DESIGN PRINCIPLES

Evidence first
Store source URL, page, clause and version.

Deterministic gates
Use code for explicit numerical rules.

Safe uncertainty
Unknown or conflicting evidence goes to review.

Resilient ingestion
Cache, deduplicate and limit crawl depth.

5. TinyFish API Integration Plan

Use each API for its intended role; keep qualification logic inside TenderLens.

STAGE 1

TinyFish Search

Discover tenders
URLs + metadata

STAGE 2

TinyFish Fetch

Retrieve pages and accessible docs
Find linked ATCs / corrigenda

STAGE 3

TinyFish Agent

Selective browser interaction
Supported workflows only

STAGE 1 · $0.00 / CALL

TinySearch API

Queries location:"IN" for live GeM, BHEL, ICMR, and PSU PDFs.

STAGE 2 · $0.00 / CALL

TinyFetch Recursive

Extracts bilingual Markdown plus links:true for child ATC files.

STAGE 3 · $0.016 / STEP

TinyAgent Stealth SSE

Selective fallback for gated GeM search filters with live SSE stream.

URL queue

Deduplicate + depth limit

PDF extraction

PyMuPDF / Docling + page refs

Verification

Python rules + LLM

Scorecard

Risk + evidence + review status

Fallback: manual upload / review if access fails. Do not assume CAPTCHA or access restrictions can be bypassed.

6. Scalability, Feasibility & Use Case

Start with an auditable MVP; expand after measuring reliability.

Monetization

Freemium: 3 tender audits/mo

MSME Pro: Monthly SaaS tier

Enterprise: Consultant API seats

Scalability

Zero-Cost Core: TinySearch/Fetch

TTL Caching: Versioned PDFs

Delta Alerts: Corrigenda watch

GTM Beachhead

Phase 1: Haryana (6,155 tenders)

Sectors: Electrical & IT MSMEs

Benchmark: Labeled GeM dataset

Risk Controls

Hard Gates: Zero math hallucination

Portal Limits: Rate-limit compliance

Audit Trail: Exact clause citations

FEASIBILITY

- Established web and document-processing stack
- Python handles explicit numerical rules
- AI assists with complex textual clauses

SCALABILITY

- Cache versioned documents
- Process audits asynchronously
- Expand portal / sector coverage incrementally

EXAMPLE USE CASE

- Contractor enters its bidder profile
- System checks turnover and certificates
- Potential gaps include page / clause evidence

Business model: Freemium audits → MSME Pro subscription → Consultant / enterprise access

7. Expected Results & Outcomes

TenderLens aims to deliver measurable impact for MSMEs, contractors and bid consultants by making tender eligibility verification faster, smarter and more reliable.

Fewer missed requirements

By analyzing the main tender document, linked ATCs and corrigenda, TenderLens ensures all eligibility conditions and amendments are considered.

- Complete document coverage (main PDF + ATCs + corrigenda)
- Automatic detection of hidden mandatory requirements
- Reduced risk of overlooking key conditions

Faster first-pass screening

AI-assisted extraction and rule-based checks help businesses quickly identify eligible opportunities and filter out high-risk tenders.

- Automated document processing and clause analysis
- Instant eligibility check against bidder profile
- Shorter time to reach a Bid / No-Bid decision

Traceable recommendations

Every finding is backed by source documents, page numbers and relevant clauses, giving users confidence in the results.

- Evidence-backed disqualification reasons
- Links to exact document pages and clauses
- Clear explanation of risks and conditions

Better Bid / No-Bid decisions

With verified eligibility checks and risk analysis, MSMEs and contractors can focus only on the tenders they are truly eligible for.

- Higher success rate in bid submissions
- Less wasted effort and reduced EMD exposure
- More efficient use of time and resources

Key Outcomes (Expected)

85%+ Disqualifier Precision (target)	**80%+** Disqualifier Recall (target)	**90%+** Linked ATC Capture Rate (target)	**< 5 mins** Avg. Processing Time (per tender)	**Lower** Manual Audit Cost (target)

Don't Just Find Tenders. Know Whether You Qualify.

Combining recursive TinyFish document ingestion, deterministic Python rules, and clause-linked AI verification for Indian MSMEs.

MILESTONE 01

50-tender benchmark

Build a labeled GeM evaluation set.

MILESTONE 02

Head-to-head ATC audit

Compare findings with manual review.

MILESTONE 03

MSME pilot

Measure usefulness, errors and time saved.

Prove accuracy before scaling.

Don't just find tenders. Know what needs to be verified before you bid.